$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

$f_n(x) \to f(x)$

$e^{ix} = \cos x + i \sin x$

Chapter 7: Sequences and Series of Functions

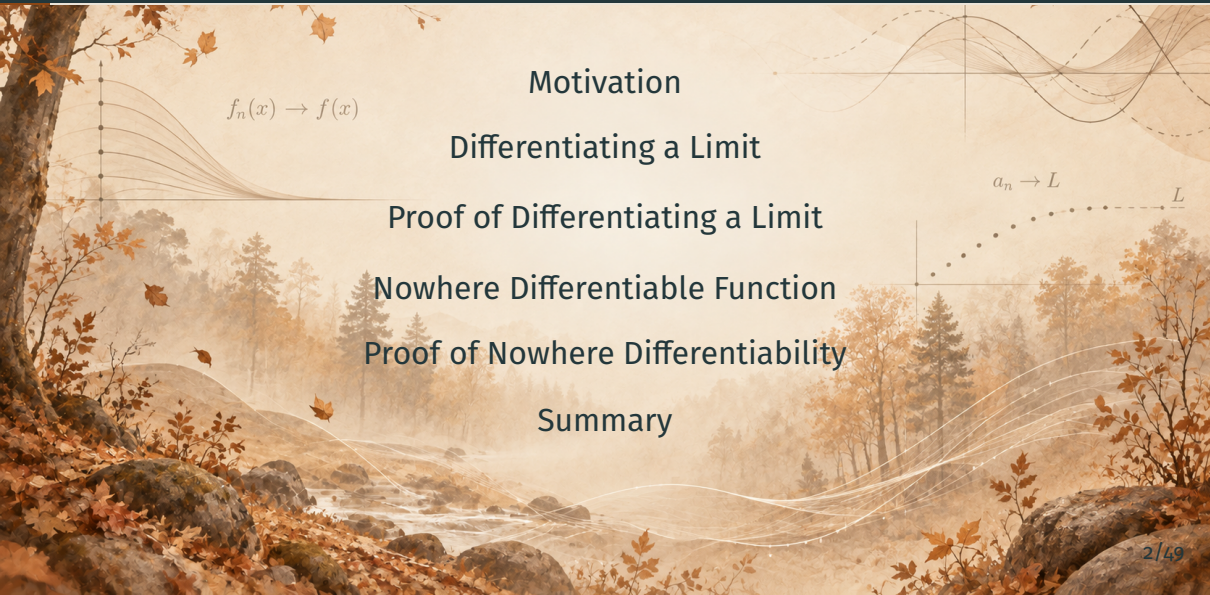
Section: Uniform Convergence and Differentiation

$a_n \to L$

Based on Walter Rudin's *Principles of Mathematical Analysis*

$S_N = \sum_{n=1}^{N} u_n$

Outline



Motivation

Differentiating a Limit

Proof of Differentiating a Limit

Nowhere Differentiable Function

Proof of Nowhere Differentiability

Summary



Motivation

Why This Section Matters

- The main problem of Section 7.1, now for derivatives:

$$\left(\lim_{n\to\infty} f_n\right)'(x) \stackrel{?}{=} \lim_{n\to\infty} f_n'(x)$$

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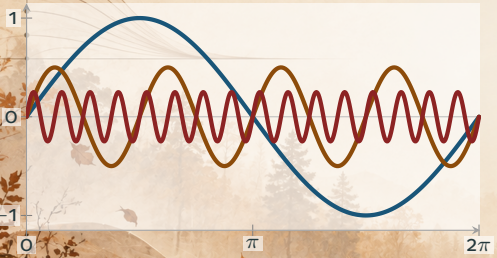
$$\left(\lim_{n\to\infty} f_n\right)'(x) \stackrel{?}{=} \lim_{n\to\infty} f_n'(x)$$

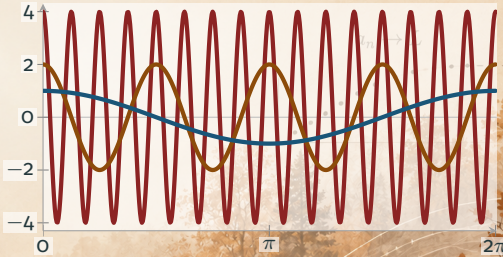
- Even *uniform* convergence $f_n \to f$ is not enough (Example 7.5)
- **Theorem 7.17:** assume instead that the derivatives f_n' converge uniformly
- **Theorem 7.18:** a uniform limit of continuous functions can be nowhere differentiable

Recall Example 7.5: Small Values, Large Slopes

$$f_n(x) = \frac{\sin nx}{\sqrt{n}} \to 0 \text{ uniformly}$$

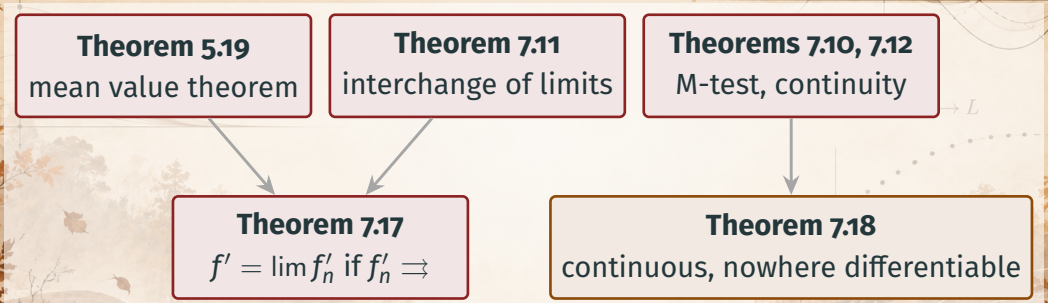
$$f_n'(x) = \sqrt{n}\cos nx \text{ does not converge}$$





$n = 1$ $n = 4$ $n = 16$ $f' \equiv 0$, but $f_n'(0) = \sqrt{n} \to +\infty$

Roadmap of This Section





Differentiating a Limit

Theorem 7.17: Differentiating a Limit

Theorem 7.17

Suppose $\{f_n\}$ is a sequence of functions, differentiable on $[a, b]$ and such that $\{f_n(x_0)\}$ converges for some point x_0 on $[a, b]$. If $\{f_n'\}$ converges uniformly on $[a, b]$, then $\{f_n\}$ converges uniformly on $[a, b]$, to a function f, and

$$f'(x) = \lim_{n\to\infty} f_n'(x) \qquad (a \le x \le b). \tag{27}$$

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In words

$$\left(\lim_{n\to\infty} f_n\right)' = \lim_{n\to\infty} f_n' \qquad \text{(differentiation and limit commute)}$$

The Role of Each Hypothesis

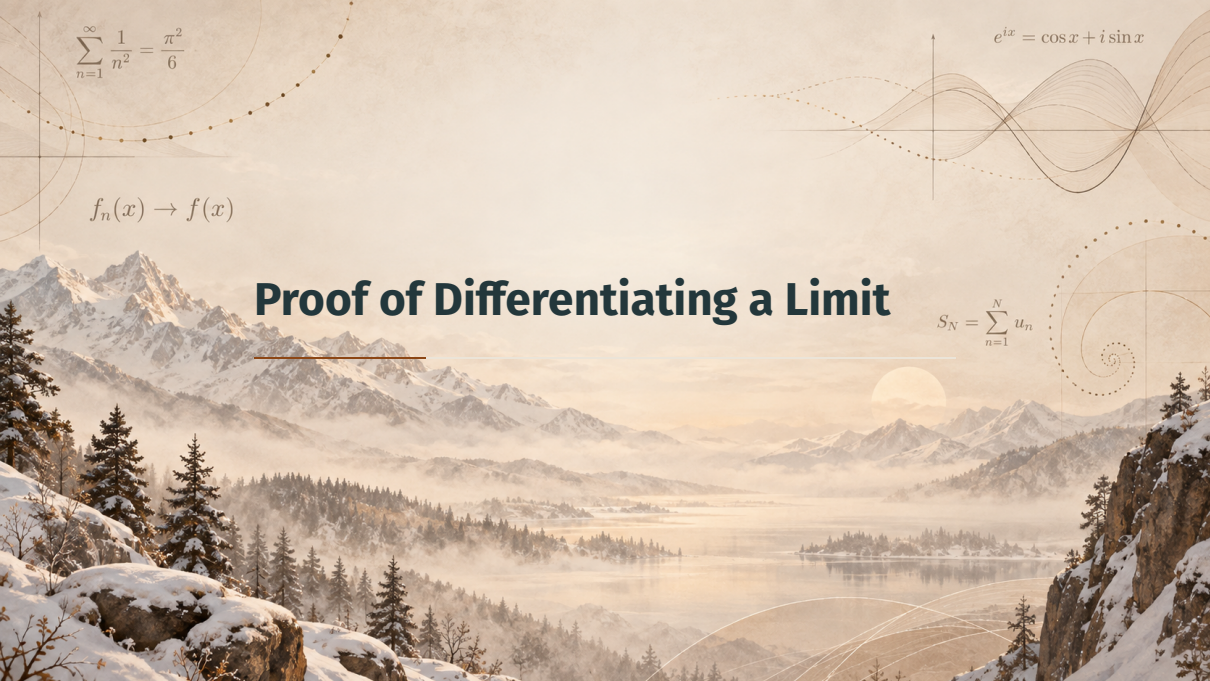
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- **Why is one point enough?**
  Slopes control *changes*: values at x_0 + uniform slopes $\Rightarrow$ values everywhere.

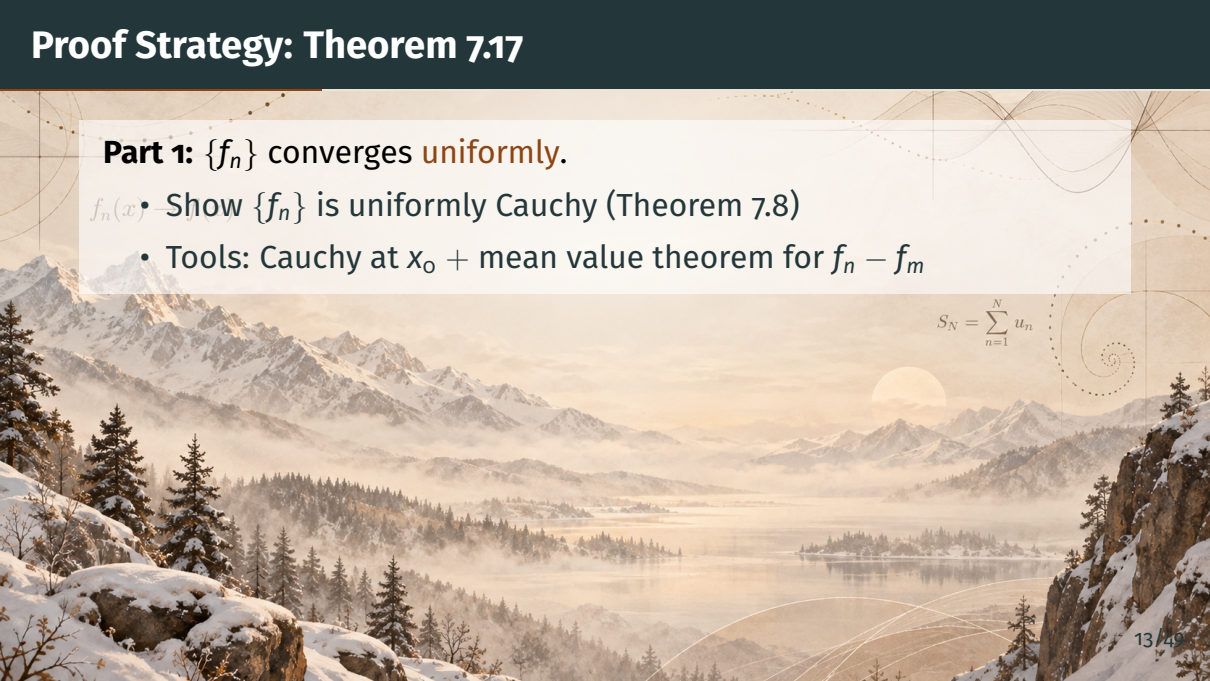


Proof of Differentiating a Limit

Proof Strategy: Theorem 7.17

Part 1: $\{f_n\}$ converges uniformly.

- Show $\{f_n\}$ is uniformly Cauchy (Theorem 7.8)
- Tools: Cauchy at x_0 + mean value theorem for $f_n - f_m$



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Part 2: $f'(x) = \lim f_n'(x)$.

- Write derivatives as limits of difference quotients
- Interchange the limits $t \to x$ and $n \to \infty$ (Theorem 7.11)

Proof of Theorem 7.17 — Step 1: Choosing N

Step 1

Let $\varepsilon > 0$. Choose N such that $n \geq N$, $m \geq N$ implies

$$|f_n(x_0) - f_m(x_0)| < \frac{\varepsilon}{2} \tag{28}$$

and

$$|f_n'(t) - f_m'(t)| < \frac{\varepsilon}{2(b-a)} \qquad (a \leq t \leq b). \tag{29}$$

(28): $\{f_n(x_0)\}$ is Cauchy. (29): $\{f_n'\}$ is uniformly Cauchy (Theorem 7.8).

Proof of Theorem 7.17 — Step 2: Mean Value Theorem

Recall: Theorem 5.19 (mean value inequality)

$|g(x) - g(t)| \leq |x - t| \cdot |g'(\xi)|$ for some ξ between x and t.

Step 2: apply it to $g = f_n - f_m$

By (29), for any $x, t \in [a, b]$ and $n, m \geq N$:

$$|f_n(x) - f_m(x) - f_n(t) + f_m(t)| \leq \frac{|x - t|\,\varepsilon}{2(b - a)} \leq \frac{\varepsilon}{2}. \tag{30}$$

Proof of Theorem 7.17 — Step 3: Uniformly Cauchy

Step 3

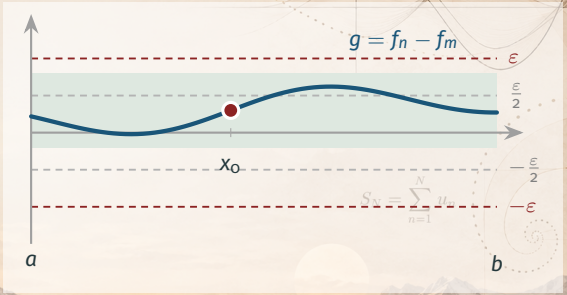
With $g = f_n - f_m$:

$$|g(x)| \le |g(x) - g(x_0)| + |g(x_0)|$$

By (30) and (28):

$$|f_n(x) - f_m(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

for all $x \in [a, b]$, $n, m \ge N$.



Proof of Theorem 7.17 — Step 3: Uniformly Cauchy

Step 3

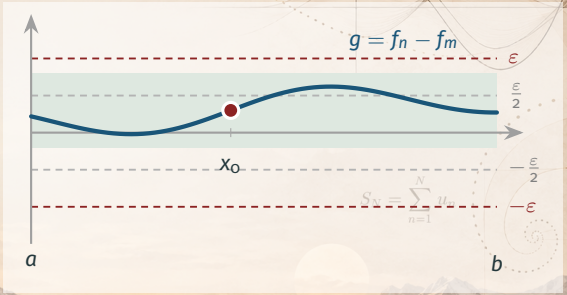
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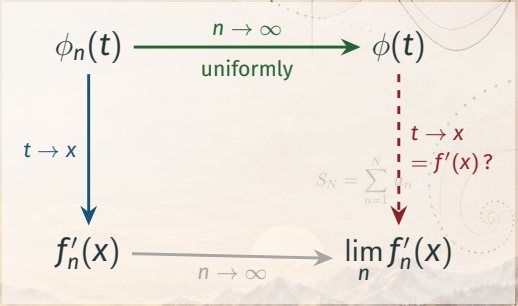
$\Rightarrow \{f_n\}$ uniformly Cauchy $\Rightarrow$ converges uniformly to some f (Thm. 7.8).

Proof of Theorem 7.17 — Part 2: Two Limits to Swap

Recall: Theorem 7.11

If $\phi_n \to \phi$ uniformly on E, x a limit point of E, and $\lim_{t\to x} \phi_n(t) = A_n$, then $\{A_n\}$ converges and

$$\lim_{t\to x} \phi(t) = \lim_{n\to\infty} A_n.$$



Proof of Theorem 7.17 — Step 4: Difference Quotients

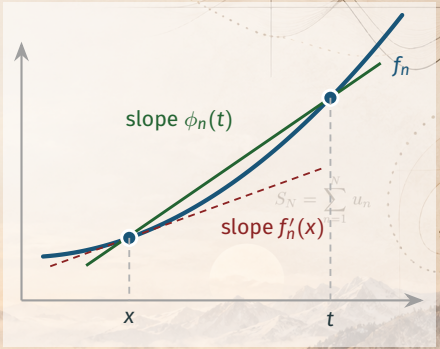
Step 4

Fix $x \in [a, b]$. For $t \in [a, b]$, $t \neq x$:

$$\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}, \qquad \phi(t) = \frac{f(t) - f(x)}{t - x}. \tag{31}$$

Then, since f_n is differentiable,

$$\lim_{t \to x} \phi_n(t) = f_n'(x). \tag{32}$$



Proof of Theorem 7.17 — Step 5: Uniform Convergence of ϕ_n

Step 5

Divide the first inequality in (30) by $|x - t|$:

$$|\phi_n(t) - \phi_m(t)| \leq \frac{\varepsilon}{2(b-a)} \qquad (n \geq N,\ m \geq N),$$

so $\{\phi_n\}$ is uniformly Cauchy, hence converges uniformly, for $t \neq x$.

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Identifying the limit

Since $f_n \to f$, equation (31) gives

$$\lim_{n\to\infty} \phi_n(t) = \phi(t) \tag{33}$$

uniformly for $a \leq t \leq b$, $t \neq x$.

Proof of Theorem 7.17 — Step 6: Conclusion

Step 6: apply Theorem 7.11 to $\{\phi_n\}$

On $E = [a, b] \setminus \{x\}$ (x is a limit point of E), by (32) and (33):

$$\lim_{t \to x} \phi(t) = \lim_{n \to \infty} f_n'(x).$$

The left side is $f'(x)$ by definition of ϕ, so

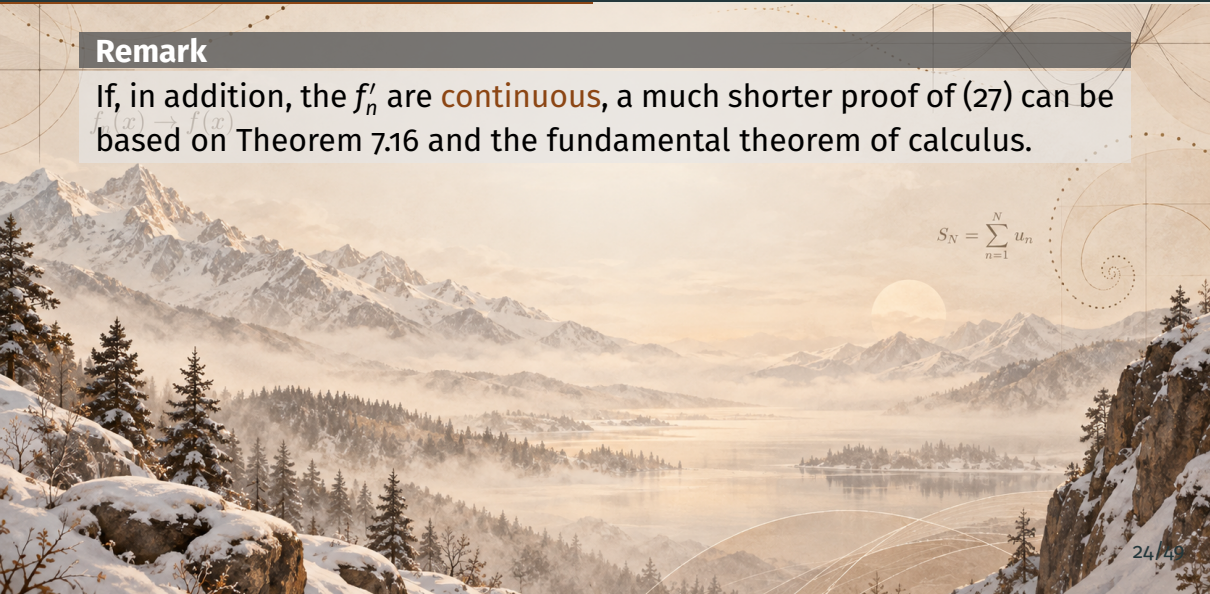
$$\boxed{f'(x) = \lim_{n \to \infty} f_n'(x)}$$

which is (27). ∎

Remark: A Shorter Proof Under Continuity

Remark

If, in addition, the f_n' are continuous, a much shorter proof of (27) can be based on Theorem 7.16 and the fundamental theorem of calculus.



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If, in addition, the f_n' are continuous, a much shorter proof of (27) can be based on Theorem 7.16 and the fundamental theorem of calculus.

Sketch

Let $g = \lim f_n'$ (continuous by Theorem 7.12). By Theorem 6.21:

$$f_n(x) = f_n(x_0) + \int_{x_0}^{x} f_n'(t)\,dt \xrightarrow[\text{Thm 7.16}]{n\to\infty} f(x) = f(x_0) + \int_{x_0}^{x} g(t)\,dt.$$

By Theorem 6.20, $f' = g$ (as g is continuous).

Consequence: Term-by-Term Differentiation

Corollary (series form of Theorem 7.17)

If each u_k is differentiable on $[a, b]$, $\sum u_k(x_0)$ converges, and $\sum u_k'$ converges uniformly on $[a, b]$, then

$$\left(\sum_{k=1}^{\infty} u_k\right)' = \sum_{k=1}^{\infty} u_k' \qquad \text{on } [a, b].$$

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Example

$$\left(\sum_{k=1}^{\infty} \frac{\sin kx}{k^3}\right)' = \sum_{k=1}^{\infty} \frac{\cos kx}{k^2}, \quad \text{since } \left|\frac{\cos kx}{k^2}\right| \le \frac{1}{k^2} \text{ and } \sum \frac{1}{k^2} < \infty \text{ (M-test).}$$

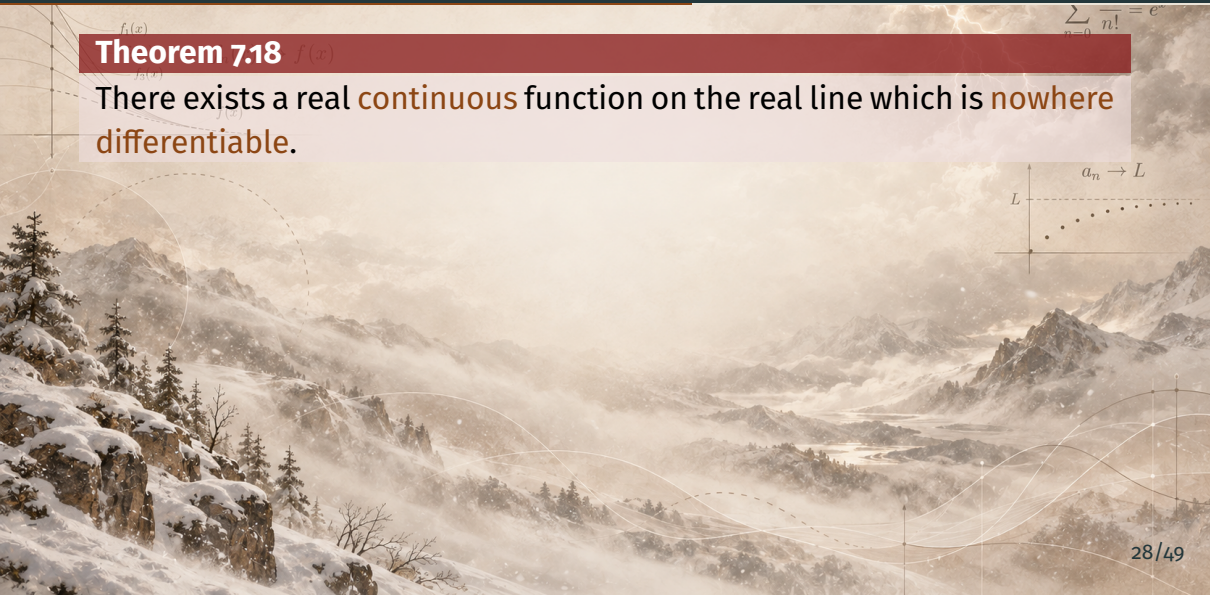


Nowhere Differentiable Function

Theorem 7.18: A Nowhere Differentiable Function

Theorem 7.18

There exists a real continuous function on the real line which is nowhere differentiable.



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There exists a real continuous function on the real line which is nowhere differentiable.

Context

- Weierstrass (1872): $\sum a^n \cos(b^n \pi x)$ caused a sensation
- Rudin's version uses a *sawtooth* instead of cosine, in the spirit of van der Waerden (1930)
- Moral: uniform limits of nice functions can be wild; Theorem 7.17 needs its strong hypothesis

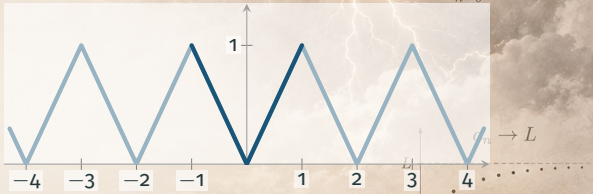
Theorem 7.18 — The Building Block φ

The sawtooth

$$\varphi(x) = |x| \quad (-1 \le x \le 1), \tag{34}$$

extended to all real x by

$$\varphi(x+2) = \varphi(x). \tag{35}$$



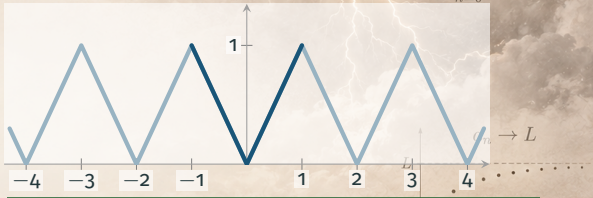
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Key properties

For all real s and t:

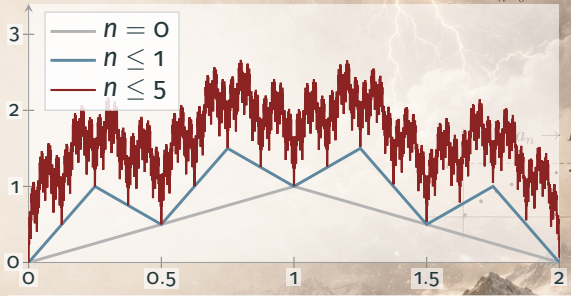
$$|\varphi(s) - \varphi(t)| \le |s - t|, \tag{36}$$

$\Rightarrow \varphi$ continuous on R^1; $0 \le \varphi \le 1$.

Theorem 7.18 — The Function f

Definition

$$f(x) = \sum_{n=0}^{\infty} \left(\tfrac{3}{4}\right)^n \varphi(4^n x). \quad (37)$$



Term n: height $\left(\tfrac{3}{4}\right)^n$ shrinks, slopes $\pm 3^n$ grow.

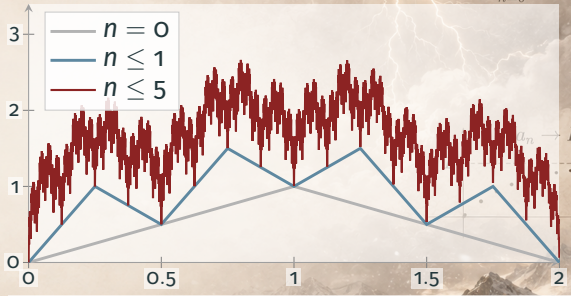
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f is continuous

$0 \le \varphi \le 1$, $\sum (\tfrac{3}{4})^n < \infty$
$\Rightarrow$ uniform (Thm. 7.10)
$\Rightarrow$ continuous (Thm. 7.12)



Term n: height $\left(\tfrac{3}{4}\right)^n$ shrinks, slopes $\pm 3^n$ grow.

$f_n(x) \to f(x)$
$(n \to \infty)$



Proof of Nowhere Differentiability

$S_N = \sum_{n=1}^{N} u_n$



Proof Strategy: Theorem 7.18

Goal: Fix any x. Show f is not differentiable at x.

Approach:

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2. Terms with $n > m$ vanish; terms with $n < m$ are *small*
3. The term $n = m$ dominates: quotient $\geq \frac{1}{2}(3^m + 1) \to \infty$

Proof of Theorem 7.18 — Step 1: Choosing δ_m

Step 1

Fix $x \in R^1$ and a positive integer m. Put

$$\delta_m = \pm\tfrac{1}{2} \cdot 4^{-m} \tag{38}$$

where the sign is chosen so that
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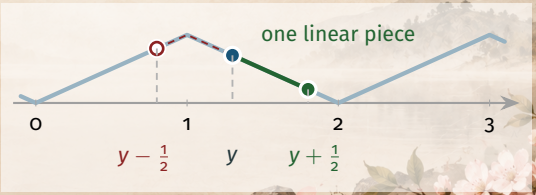
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Possible since $4^m|\delta_m| = \frac{1}{2}$: one half-step from $y = 4^m x$ stays in the unit interval of y.



Proof of Theorem 7.18 — Step 2: The Quotients γ_n

Step 2

Define

$$\gamma_n = \frac{\varphi(4^n(x+\delta_m)) - \varphi(4^n x)}{\delta_m}. \tag{39}$$

Then, by (37), $\dfrac{f(x+\delta_m)-f(x)}{\delta_m} = \displaystyle\sum_{n=0}^{\infty} \left(\tfrac{3}{4}\right)^n \gamma_n.$

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- $n > m$: $4^n\delta_m = \pm 2 \cdot 4^{n-m-1}$ is even $\Rightarrow \gamma_n = 0$ (period 2)
- $0 \le n \le m$: by (36), $|\gamma_n| \le 4^n|\delta_m|/|\delta_m| = 4^n$
- $n = m$: $|\gamma_m| = 4^m$ (slope ± 1 on one piece, Step 1)

Proof of Theorem 7.18 — Step 3: The Estimate

Step 3

$$\left|\frac{f(x+\delta_m)-f(x)}{\delta_m}\right| = \left|\sum_{n=0}^{m}\left(\tfrac{3}{4}\right)^n \gamma_n\right| \geq \underbrace{\left(\tfrac{3}{4}\right)^m 4^m}_{\text{term } n=m} - \sum_{n=0}^{m-1}\left(\tfrac{3}{4}\right)^n 4^n$$

$$= 3^m - \sum_{n=0}^{m-1} 3^n = 3^m - \frac{3^m-1}{2} = \boxed{\tfrac{1}{2}(3^m+1)}.$$

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Conclusion

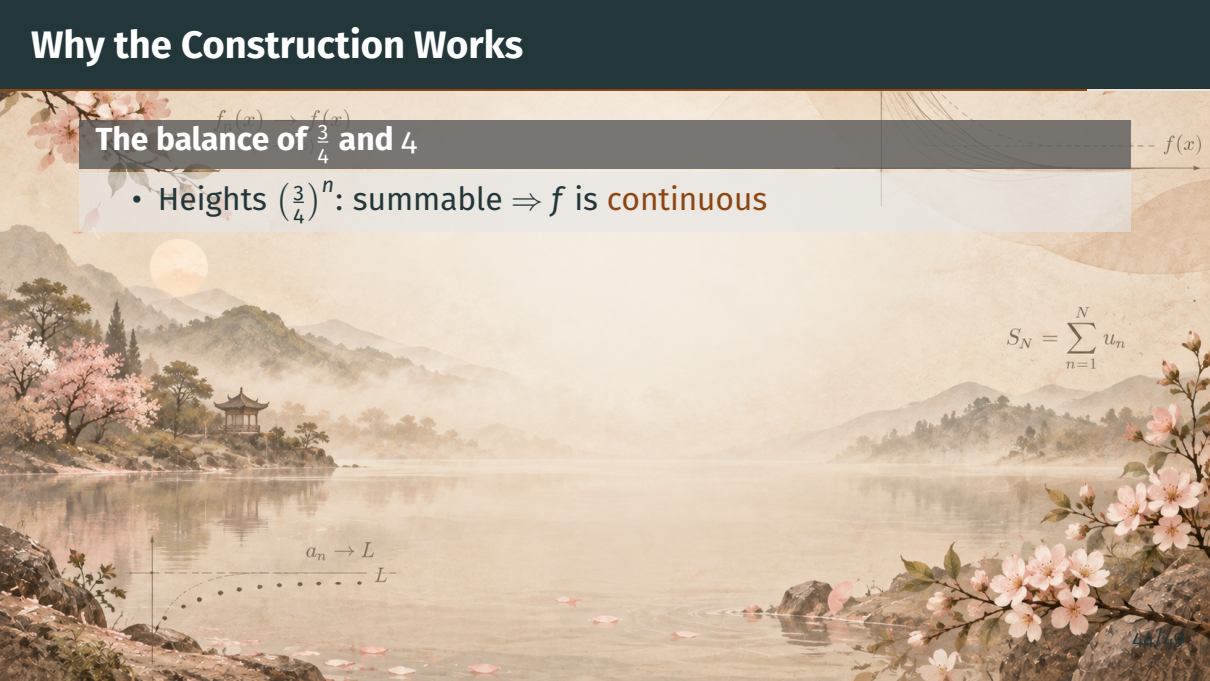
As $m \to \infty$: $\delta_m \to 0$, but the quotients $\to \infty$.

Hence f is not differentiable at x. Since x was arbitrary, f is nowhere differentiable. ∎

Why the Construction Works

The balance of $\frac{3}{4}$ and 4

- Heights $\left(\frac{3}{4}\right)^n$: summable $\Rightarrow f$ is continuous



Why the Construction Works

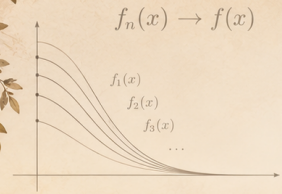
The balance of $\frac{3}{4}$ and 4

- Heights $\left(\frac{3}{4}\right)^n$: summable $\Rightarrow f$ is continuous
- Slopes $\left(\frac{3}{4} \cdot 4\right)^n = 3^n$: each one exceeds the *sum* of all earlier ones $\Rightarrow$ no derivative

Why the Construction Works

The balance of $\frac{3}{4}$ and 4

- Heights $\left(\frac{3}{4}\right)^n$: summable $\Rightarrow f$ is continuous
- Slopes $\left(\frac{3}{4}\cdot 4\right)^n = 3^n$: each one exceeds the *sum* of all earlier ones $\Rightarrow$ no derivative
- Same proof for $\sum a^n \varphi(b^n x)$ with $0 < a < 1$, $4 \mid b$, $ab > 2$: quotients $\geq (ab)^m \dfrac{ab-2}{ab-1} \to \infty$



$S_N = \sum_{n=1}^{N} u_n$



$\lim_{n \to \infty} u_n = L$

Summary



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- **Theorem 7.18:** $\sum (\frac{3}{4})^n \varphi(4^n x)$ is continuous but nowhere differentiable

Coming up next: equicontinuous families of functions.